THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bicycle wheel is spun in the air; it stops after a while without brakes. This is due to: 1

(a) friction at the axle and air resistance **(b)** gravity pulling it horizontally
(c) its mass **(d)** magnetic force

2. Assertion (A): Two like magnetic poles attract each other. Reason (R): Magnetic poles always occur in pairs. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Two forces of 2 N each act on a body along the same straight line. The net force is 4 N when they act: 1

(a) on different bodies **(b)** in the same direction
(c) in opposite directions **(d)** at different times

4. Which of these will NOT stay charged after being rubbed? 1

(a) a glass rod rubbed with silk **(b)** a dry plastic comb
(c) a dry balloon **(d)** a metal rod held in the bare hand

5. Assertion (A): A spring balance can measure the weight of an object. Reason (R): Weight is a force and the spring stretch depends on force. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. An astronaut releases a hammer and a feather on the Moon. They land together because: 1

(a) gravity is zero **(b)** the feather is heavy
(c) there is no air resistance **(d)** the hammer is light

7. Which statement about weight and mass is correct? 1

(a) Weight depends on location but mass does not
(b) Mass depends on location but weight does not
(c) Both are the same
(d) Both change on the Moon

8. Assertion (A): Wet roads increase the chance of skidding. Reason (R): Water reduces friction between tyres and road. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** The weight of a body is measured in newton. 1
Reason (R): A spring balance measures the mass of a body in kilogram.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Smooth-looking surfaces still experience friction. 1
Reason (R): Even polished surfaces have tiny irregularities that interlock with each other.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the compass needle always settle in the north-south direction? 2

12. Why is a fully inflated tyre easier to roll than a flat tyre? 2

13. Why can friction not be completely removed from a machine even with good lubrication? 2

14. Why are ski and skates made with slippery bases and shoes with spikes? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A spring balance shows 50 N for a box. A second identical spring balance is used to also pull upward with 20 N on the box while it hangs from the first. What does the first balance read now? 3

16. Why does an aeroplane use less fuel at a lower speed than at a higher speed? 3

17. A carrom coin is hit on a powdered board and on an unpowdered board with the same force. Compare and explain. 3

OR

Explain why an astronaut can jump higher on the Moon than on the Earth. 3

18. A bag is pulled over rough ground by a boy and a girl in different directions. Describe how the net force can be zero or non-zero. 3

OR

Why does a charged comb attract a thin stream of water from a tap? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rehana rubs a plastic scale on dry hair and brings it near tiny pieces of paper, which jump to it. Later, in a humid monsoon evening, the same trick fails. Her teacher says that rubbing causes charging by friction, and that moist air helps the charge to leak away. 5
(i) Name the force that attracts the paper bits.
(ii) Is it a contact or non-contact force?
(iii) How does the scale get charged?
(iv) Why does the experiment fail on a humid day?

20. The mass of an object is 12 kg. Using 10 N per kg on Earth and one-sixth on the Moon, find its weight on each and compare. Explain which quantity would be measured by a beam balance and by a spring balance. 5

OR

Analyse the force of friction and the force of gravity on a skier going down a slope. Describe how the forces change at the bottom of the slope and why the skier stops. 5

END OF QUESTION PAPER — Total: 40 Marks